Guide

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Introduction

Use this interactive app to explore the logic of hypothesis testing through visual and dynamic experimentation. Modify the parameters and observe how the p-value, the critical region and the test decision change.

The simulator contains four panels:

- p-value
- Test on a coefficient
- Test on the whole model
- Test on a subset of parameters

Work through them actively. Change parameters, observe the graphs and interpret the results.

General structure of the app

The interface has two main areas.

On the left, use the parameter panel to modify the inputs of each test. You can change the significance level, observed statistics, sample size and model characteristics.

On the right, observe the results. Each tab displays:

- The graph of the sampling distribution
- The rejection region
- The p-value
- The decision of the test
- An interpretation of the result

Although each tab represents a different statistical situation, all follow the same decision logic.

Tab 1. The concept of the p-value

Start by exploring the graphical meaning of the p-value.

Observe the curve representing the distribution of the test statistic under the null hypothesis. The vertical line shows the observed value from the sample. The red shaded area represents the p-value. The blue region represents the rejection region determined by the significance level.

Modify the observed statistic and the significance level. After each change, answer:

- Does the p-value increase or decrease?
- Does the rejection region change?
- Is the null hypothesis rejected?

Use this tab to understand what the p-value represents and how the decision rule works.

Tab 2. Test on a coefficient

Move to the regression coefficient test.

Modify the following parameters:

- Estimated coefficient
- Standard error
- Sample size
- Number of explanatory variables
- Value under the null hypothesis

Observe how the t statistic, p-value and decision change. After each modification, interpret the result.

In particular, explore the following:

- Increase and decrease the standard error while keeping the coefficient fixed.
- Modify the sample size and the number of explanatory variables.
- Change the value of the null hypothesis from zero to another number.
- Compare cases where the coefficient is statistically significant and not significant.
- Use the graph to connect formulas, output and interpretation.

Tab 3. Test on the whole model

Use this tab to analyse the overall significance of a regression model.

Modify:

- R^2 of the model
- Sample size
- Number of explanatory variables
- Significance level

Observe how the F statistic changes and how this affects the p-value and the decision.

Check how:

- A higher R^2 affects significance
- A larger sample size affects the result
- The rejection region depends on the significance level

Interpret whether the model is globally significant.

Tab 4. Test on a subset of parameters

Use this tab to analyse joint restrictions on a subset of coefficients.

Modify:

- R^2 of the full model
- R^2 of the restricted model
- Sample size
- Number of explanatory variables
- Number of restrictions

Observe how the difference between the full and restricted model affects the F statistic and the decision. Try several combinations and determine when the restrictions are rejected.